

Harshit Mishra

9628492644 | harshithihu@gmail.com | Portfolio | LinkedIn | GitHub

EDUCATION

Indian Institute of Information Technology, Senapati, Manipur

Bachelor of Technology in Computer Science and Engineering

Manipur, India

Aug 2023 – May 2027

- **CGPA: 7.02/10.0** | Coursework: Data Structures & Algorithms, OS, OOP, DBMS, Compiler Design

L.D. Convent School

UP Board (Percentage: 83.8%)

Uttar Pradesh, India

2022

EXPERIENCE

Varun Beverages Limited

Data Analyst Intern

20 May 2026 – 13 Jul 2026

- Developed a Market Basket Analysis solution using Apriori and FP-Growth algorithms to identify frequently purchased product combinations and generate cross-selling recommendations.
- Built an end-to-end Customer Churn Prediction pipeline using Logistic Regression, Random Forest, and XGBoost with feature engineering, hyperparameter tuning, and model evaluation.
- Performed data cleaning, exploratory data analysis (EDA), visualization, and deployed interactive Streamlit dashboards to deliver actionable business insights.

PROJECTS

OmniScan AI: Autonomous Vulnerability Scanner | *Python, Node.js, React, RAG, Ollama*

2026

- Built a 100% local, privacy-centric cybersecurity scanner with custom SAST/DAST engines in Node.js and a React frontend.
- Integrated a Python-based RAG pipeline with FAISS vector database to provide OWASP contextual knowledge to a local Phi-3 LLM via Ollama.
- Implemented **Attacker Personas** and vulnerability chaining to simulate multi-step cyberattacks beyond traditional scanners.
- Generated automated, developer-focused remediation code using LLM inference on detected vulnerability chains.

AGRIHIVE – AI Farming Advisory Platform | *Flutter, Firebase, TensorFlow, Gemini API*

Mar 2025 – Jun 2025

- Developed an end-to-end AI-powered platform for rural farmers with plant disease detection, advisory chatbot, and real-time crop health monitoring.
- Built and trained a TensorFlow CNN model for plant disease classification achieving **92% accuracy** on real-world field data.
- Integrated Gemini API for multilingual, context-aware farming assistance with natural language query support.
- Designed an offline-first Firebase backend (Firestore + Cloud Functions) supporting **100+ concurrent** rural users with low-latency data synchronization.

NovaVCS – High-Performance Version Control System | *C++17, System Architecture*

Aug 2026

- Engineered a high-performance Version Control System (VCS) from scratch in C++17, implementing a Directed Acyclic Graph (DAG) for commit history and Content-Addressable Storage (CAS).
- Developed a custom diff and merge engine utilizing the optimized O(ND) Myers algorithm and a three-way merge strategy, serializing conflict states into structured JSON for automated resolution.
- Built a sub-millisecond search engine integrating a Trie for prefix matching, an Inverted Index for content queries, and Levenshtein automata for fuzzy search capabilities.
- Implemented a local compression engine using Huffman Coding and delta compression to optimize loose object storage into efficient packfiles.
- Designed a comprehensive analytics module that parses repository metrics into JSON payloads, engineered for seamless API ingestion by a scalable PERN stack enterprise dashboard.

TECHNICAL SKILLS

Languages: Python, C++, SQL, JavaScript

Backend: REST APIs, Flask, Firebase, Node.js, Docker

Data Analytics: Pandas, NumPy, Matplotlib, EDA, ETL Pipelines, Statistical Analysis

ML / AI: TensorFlow, CNN, scikit-learn, XGBoost, NLP, LLMs, Prompt Engineering

Tools: Git, Linux, Postman, Jupyter Notebook, Streamlit

ACHIEVEMENTS

LeetCode | *Competitive Programming*

2025

- Achieved a **Global Contest Rank of 1623** on LeetCode, demonstrating strong problem-solving skills under timed contests.
- Solved problems across Arrays, Graphs, Dynamic Programming, Binary Search, Trees, and Greedy algorithms with consistent practice on medium-to-hard difficulty questions.